C Evaluation on Additional Models and Datasets

We include additional alignment results in Figures 19–22 using a broader set of sensory encoders extended to additional datasets. For vision, we evaluate DINOv2 [Oquab et al., 2023], CLIP [Radford et al., 2021], ViT-MAE [He et al., 2022], and ViT-MSN [Assran et al., 2023]. For audio, we evaluate CLAP [Elizalde et al., 2023], BEATs [Chen et al., 2022], BEATs+ (BEATs fine-tuned on AudioSet labels), Audio-MAE [Huang et al., 2022], and EAT [Gong et al., 2023]. These models are compared across WiT (image–text), AudioCaps (audio–text), DCI (image–text), and Clotho (audio–text) datasets.

Encoders supervised on language such as CLIP (vision–language) and CLAP (audio–language) show the strongest alignment with LLM representations across all prompt types, reflecting their direct training on paired data. Self-supervised models such as DINOv2, ViT-MAE, ViT-MSN, BEATs, Audio-MAE, and EAT exhibit weaker but still consistent alignment trends. BEATs+ shows higher alignment over its self-supervised counterpart, which highlights the role of additional semantic supervision in shaping compatibility with text-trained LLMs.

We also report token–alignment trends for additional language models (Figures 23–25). We include Meta's Llama 3/3.1 family [Grattafiori et al., 2024; Meta AI, 2024] and Microsoft's Phi-4 [Abdin et al., 2024], alongside our Qwen3 baselines.

These results demonstrate that our sensory-prompting findings are not tied to any specific encoder, dataset, or language model: alignment effects generalize across self-supervised and multimodally supervised encoders and across multiple LLM families (Llama 3/3.1, Phi-4, Qwen3). Moreover, the strongest alignment consistently arises from encoders with explicit multimodal supervision.

D Visual Bias in Auditory Setting

To further understand how sensory prompts shape internal representations, we compute pairwise similarity between no cue, SEE, and HEAR generations using both centered kernel alignment (CKA) and mutual-kNN alignment (k=10) (Figure 26).

Across most scales, no cue embeddings are consistently closer to SEE than HEAR, even on the auditory AudioCaps dataset—demonstrating a persistent visual inductive bias under default prompting. To address the possibility that the instruction "imagine" may implicitly induce visual imagery, we confirm that the same trend holds under "describe" instructions (Appendix Figure 28), suggesting that auditory structure is weakly activated by default and that explicit HEAR cues greatly improve alignment with audio encoders. At 32B scale, no cue diverges from both SEE and HEAR on AudioCaps. While this may reflect a shift toward a more modality-agnostic default representations in larger models, we view this as a preliminary observation. Interestingly, this difference is less stark using "describe" instructed prompting.

We extend Figure 26 to "describe" instructed prompting in Figure 28 and to the DCI dataset in Figure 27.

E Extended Analysis of Sensory Axis Projections

To complement the qualitative distribution plots in Figure 8 (a), we quantify SEE–HEAR separation along the learned projection axis in Table 2. We report three metrics: $\Delta\mu$, the raw difference between the mean SEE and HEAR projections (larger values indicate greater directional shift); Cohen's d, the standardized effect size that rescales $\Delta\mu$ by within-class variance ; and AUROC, a rank-based discriminability score reflecting how well a single threshold separates SEE from HEAR (0.5 = chance, 1.0 = perfect).

Figure 29 extends our projection analysis (Figure 8 (a)) to the DCI dataset. Compared to WiT, DCI shows stronger disentanglement between SEE and HEAR cues—reflected in wider separation of the projection distributions. We hypothesize this is because DCI captions contain richer inherently visual detail, such as textures, layouts, and scene composition, whereas WiT captions often reference proper nouns, locations, or events like "Finster/Nagy at the 2019 World Junior Championships" or "Unnamed hurricane of 1975 near the Pacific Northwest", which offer less explicit sensory content and may reduce the contrast in projection space.

Figure 30 shows projections under "describe" prompting, which we compare to "imagine" prompting from Figure 8

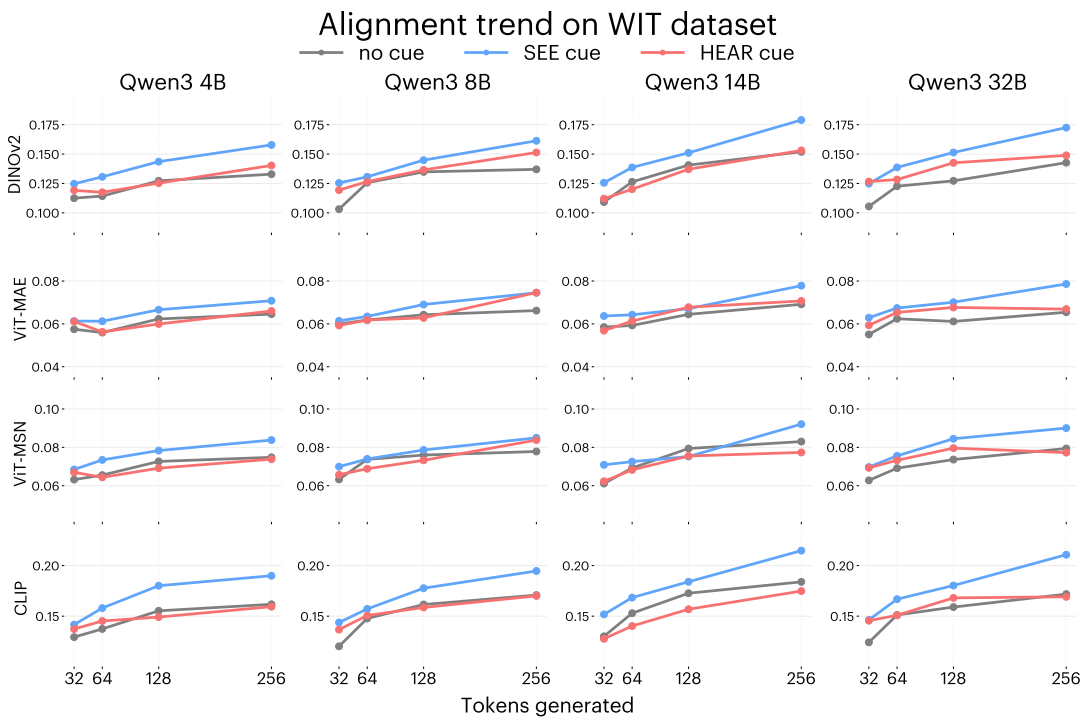

Figure 19: Extension of Figure 6 to additional sensory encoders on WIT.

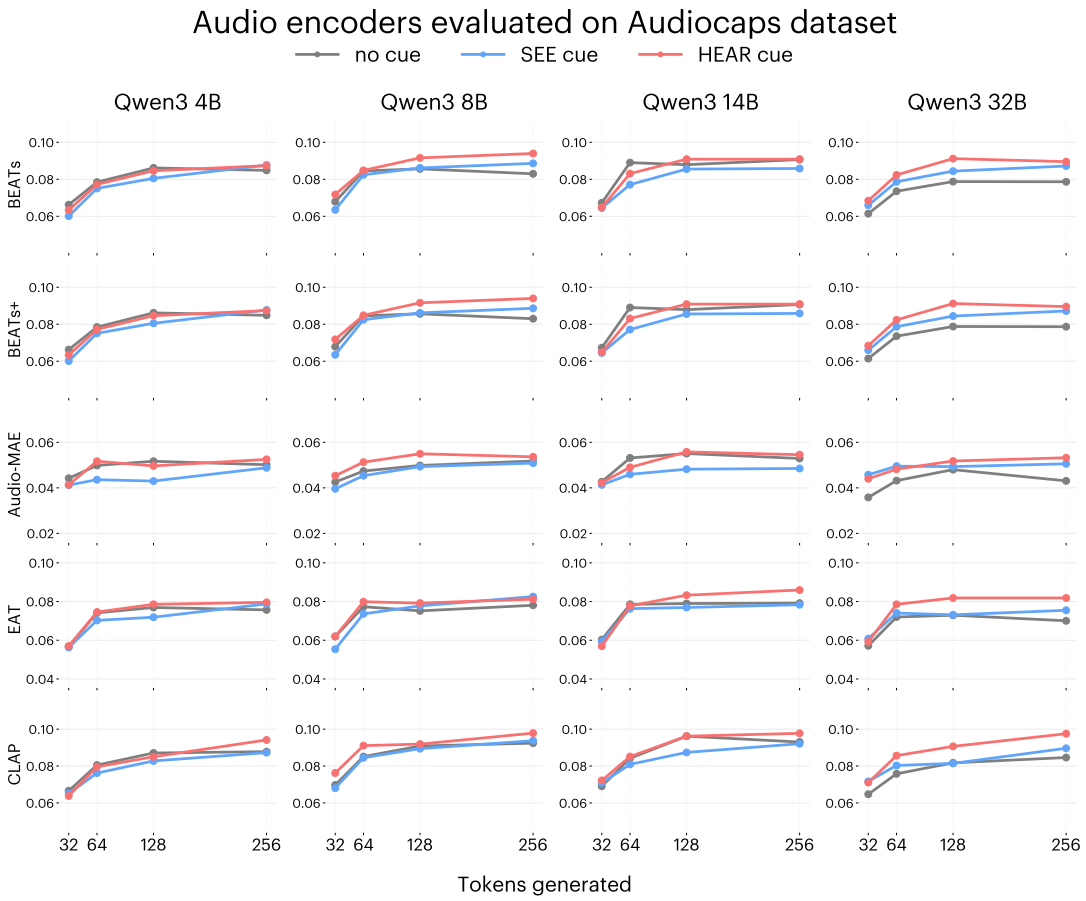

Figure 20: Extension of Figure 6 to additional sensory encoders on AudioCaps.

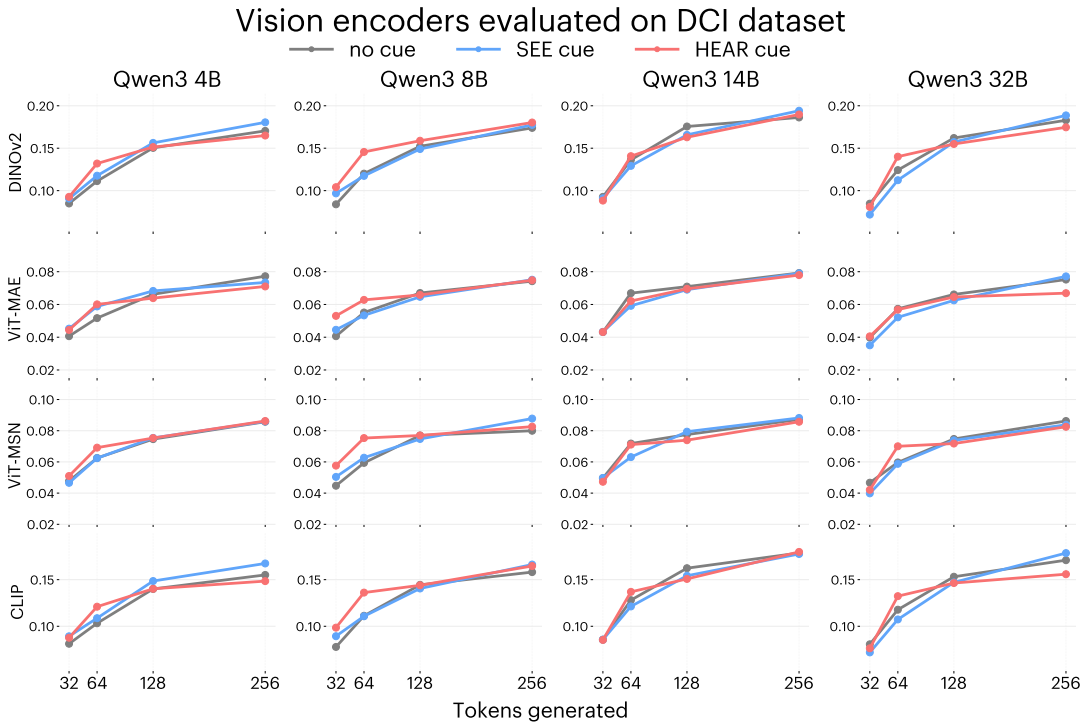

Figure 21: Extension of Figure 6 to additional sensory encoders on DCI.

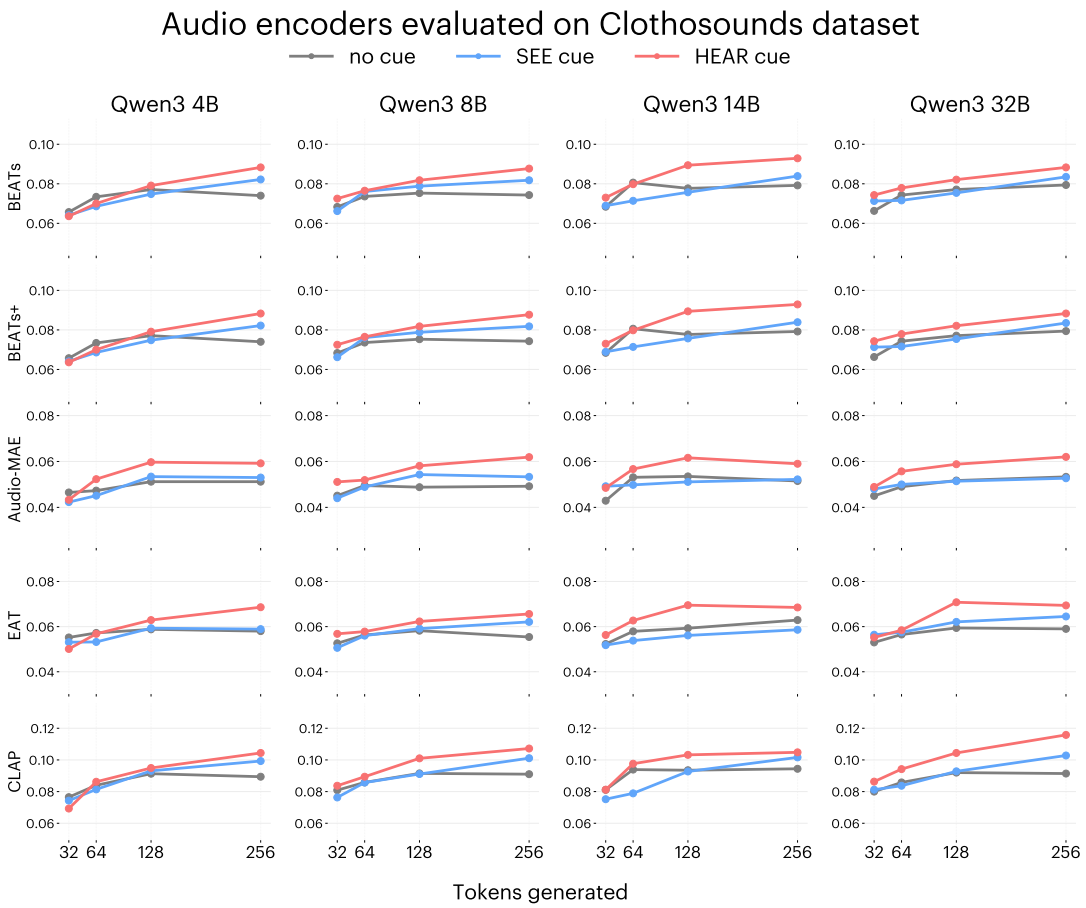

Figure 22: Extension of Figure 6 to additional sensory encoders on Clothosounds.